🔑 **Answer Key — Teacher Copy · Lesson 8-4**

MSTAR PRACTICE WORKSHEET

Lesson 8-4: Write and Represent Inequalities

Grade 6 Mathematics · Standard 6.AT.9

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'The temperature is less than 40 degrees'?

- **A** t < 40
- **B** t > 40
- **C** t ≤ 40
- **D** t = 40

'Less than' means <, so the inequality is t < 40.

Question 2 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'You must be at least 48 inches tall to ride'?

- **A** h ≥ 48
- **B** h > 48
- **C** h < 48
- **D** h ≤ 48

'At least' means greater than or equal to, so h ≥ 48.

Question 3 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'A store can hold no more than 75 people'?

A $p \leq 75$

B $p < 75$

C $p \geq 75$

D $p > 75$

'No more than' means less than or equal to, so $p \leq 75$.

Question 4 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'The speed limit is more than 25 mph'?

A $s > 25$

B $s \geq 25$

C $s < 25$

D $s = 25$

'More than' means strictly greater than, so $s > 25$.

Question 5 SELECTED RESPONSE — CORRECT: A

A parking garage charges if your stay is more than 2 hours. Which inequality represents times t that are charged?

A $t > 2$

B $t \geq 2$

C $t < 2$

D $t \leq 2$

'More than 2 hours' means strictly greater than: $t > 2$. Exactly 2 hours is not charged.

Question 6 SELECTED RESPONSE — CORRECT: A

A clue says the suspect carried at least 12 stolen gems. Which inequality shows the number g of gems?

- (A) $g \geq 12$
- (B) $g > 12$
- (C) $g \leq 12$
- (D) $g < 12$

'At least 12' includes 12 and any number above it, so $g \geq 12$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A fair ride requires each rider to be at least 48 inches tall. Let h be a rider's height in inches. Which inequality represents the heights that are allowed to ride?

(A) $h \geq 48$

(B) $h > 48$

(C) $h \leq 48$

(D) $h < 48$

> The phrase 'at least 48 inches' means 48 inches is allowed and any greater height is allowed too. That is greater than or equal to, so the inequality is $h \geq 48$.

- **B:** This shuts out a rider who is exactly 48 inches tall. 'At least' includes the 48.
- **C:** This admits only riders 48 inches and SHORTER. Taller riders are allowed too.
- **D:** This admits only riders under 48 inches, which is the opposite of the rule.

Part B — correct answer: C

Which explanation justifies the symbol chosen above?

(A) The words 'at least' mean a rider must be taller than 48 inches, so 48 inches is not allowed.

(B) The words 'at least' point to the smallest numbers, so riders must be shorter than 48 inches.

(C) The words 'at least' include the boundary value, so a rider who is exactly 48 inches tall may ride.

(D) The words 'at least' and 'at most' describe the same group of heights, so either symbol works.

> 'At least' sets a minimum that is part of the group, which is why the symbol carries the bar underneath. A rider who measures exactly 48 inches meets the rule, so $h \geq 48$ rather than $h > 48$.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Reversing 'No More Than'

> Priya read this situation: 'A backpack can hold no more than 15 pounds.' She let w stand for the number of pounds in the backpack and wrote the inequality $w \geq 15$.

Explain Priya's misconception and give the correct inequality.

Model answer: $w \leq 15$. 'No more than 15 pounds' means the weight may be 15 pounds or anything below 15 pounds, so the correct inequality is $w \leq 15$. Priya's inequality $w \geq 15$ describes weights of 15 pounds and heavier, which is the opposite group. She kept the boundary number but pointed the symbol the wrong way.

2 points	Student explains that 'no more than' sets a maximum the weight cannot pass, so the symbol must be less than or equal to, and writes $w \leq 15$.
1 point	Student writes $w \leq 15$ but does not explain why 'no more than' sets a maximum.
0 points	Student agrees with Priya or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Write two different real-world situations: one that uses > and one that uses ≥. Explain why the situations need different symbols even though both mean 'bigger.'

Model answer: The only difference between > and ≥ is whether the boundary number itself is allowed. A rule like "score more than 10 points to win" leaves 10 out, so it needs >, while "you must be at least 13 to watch" lets exactly 13 count, so it needs ≥. Testing the boundary number in your own situation is the quickest way to check which symbol fits.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.